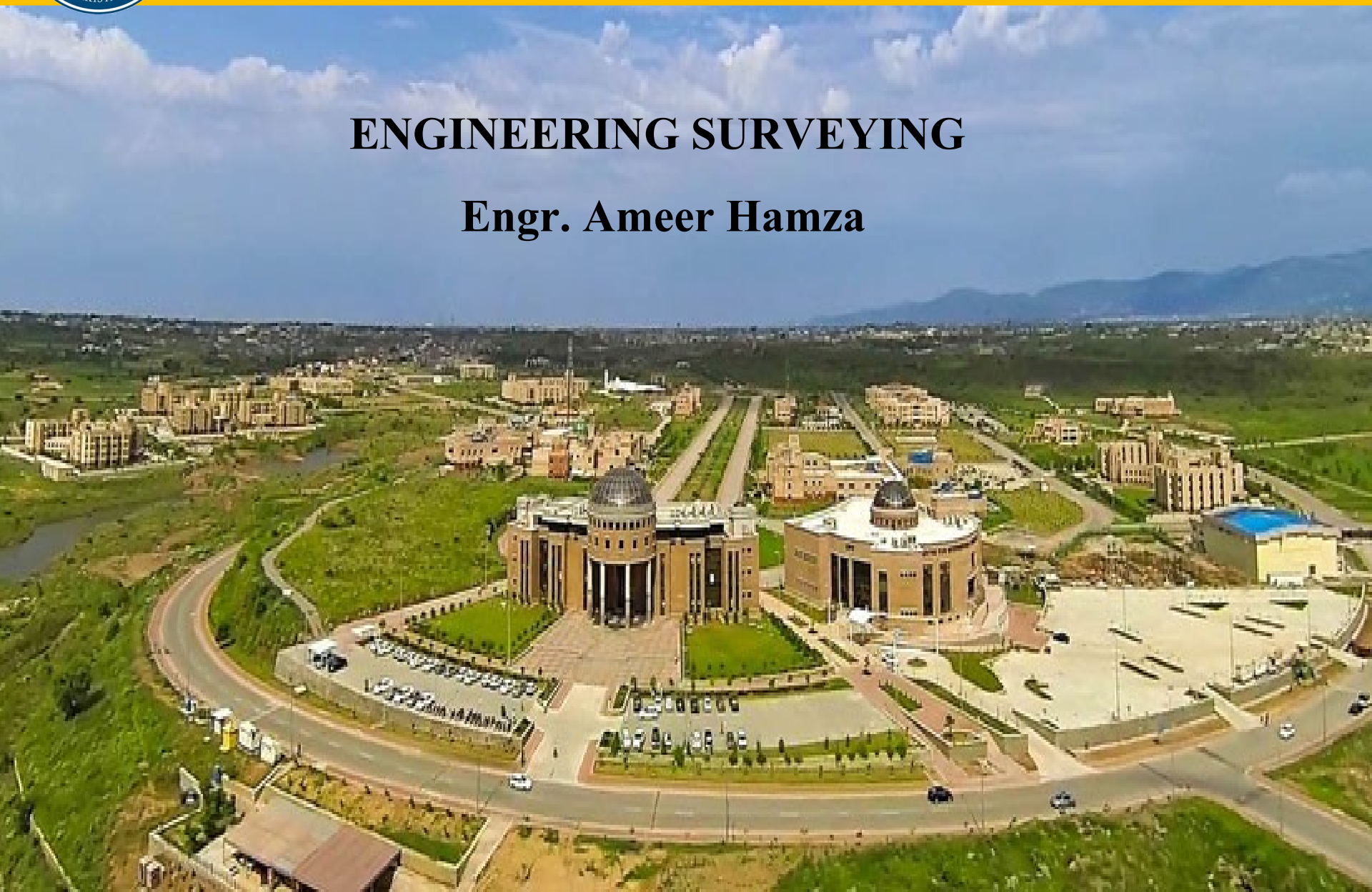

بسم الله الرحمن الرحيم

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ENGINEERING SURVEYING

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INTRODUCTION
To
ENGINEERING
SURVEY

DEFINITION

SURVEY

- The science of measurement with the objective of increasing our knowledge about shape of the earth and features upon it.
- **Plane survey** refers to survey of small extent where it is assumed that the mean surface of the earth is flat for the area concerned.
- **Geodetic survey** is for larger areas where this assumption is not true and allowance must be made for the curvature of the earth.

HOW FAR YOU CAN SEE BEFORE THE CURVATURE OF EARTH IMPACTS YOUR LINE OF SIGHT.

Radius of earth (R)= 6372797 meters

Height of observer(H) = 2m

Calculations:-

$(R+H)^2 = X^2 + R^2$

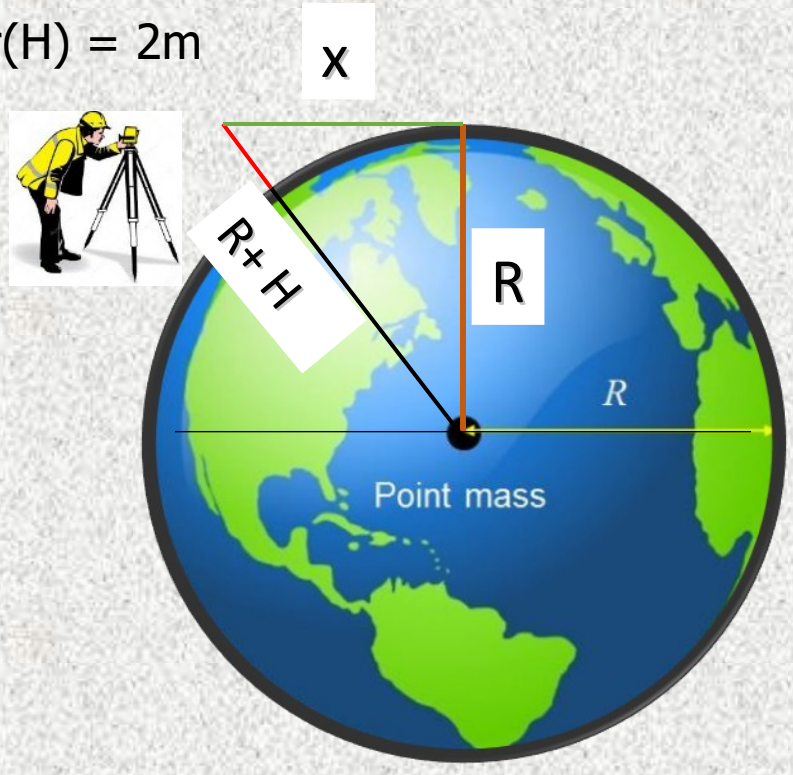
$X^2 = (R+H)^2 - R^2$

$X^2 = (6372797+2)^2 - (6372797)^2$

$X^2 = (6372799)^2 - (6372797)^2$

X = 5048.88m

X = 5.04km



TYPES OF SURVEY

1. Geodetic Survey

- It is the process of making measurement with a view to determine the size and shape of the earth. It provides the main framework within which all topographical, cadastral and engineering surveys can be located.

2. Topographical Survey

- Topographical survey is concerned with the measurement of natural and artificial features of the earth's surface so that a map of these features may be drawn and printed.

TYPES OF SURVEY

3. Cadastral Surveying

- It is the process of defining, demarcating, measuring and recording the boundaries of properties. Where these boundaries are formed by physical features, it overlaps with topographical surveying. As a general rule, cadastral work is done at larger scales than topographical mapping

4. Engineering Surveys

- Engineering Surveys are those conducted with the special object of supplying particular information for engineering projects. They are usually at larger scales than topographical maps, but the methods used are often similar. Sometimes a high order of accuracy is required, for example, the measuring of a gap for a bridge.

TYPES OF SURVEY

5. Topographic Map

- The topographical map is one which shows all the information about natural and artificial objects. Scales vary between 1/5000 to 1/1000000 depending on the purpose for which the map is made.

PRINCIPALS OF SURVEY

1. CONTROL

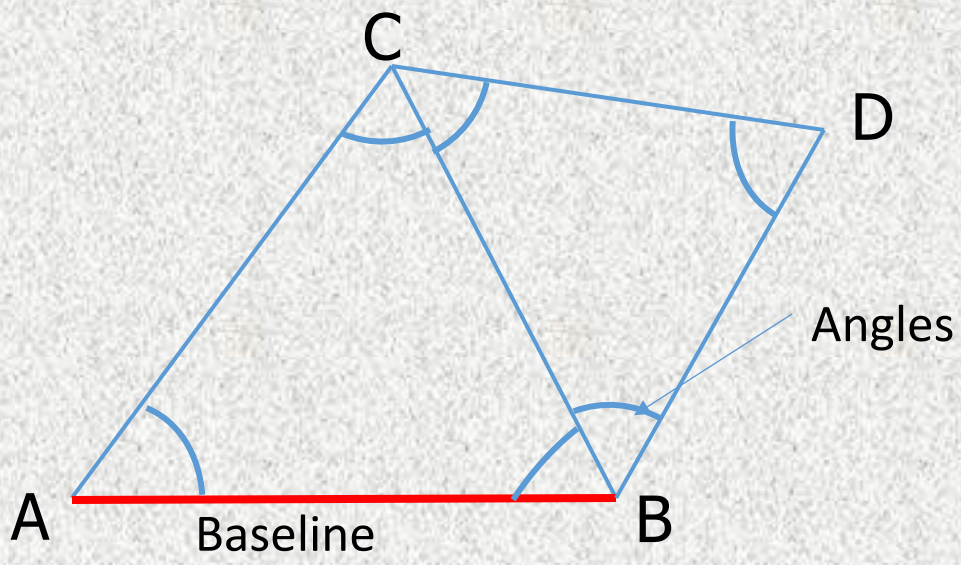
- Framework encompassing entire area
- Free of error
- Subsequent work adjusted to it
- Working from whole to part

METHODS

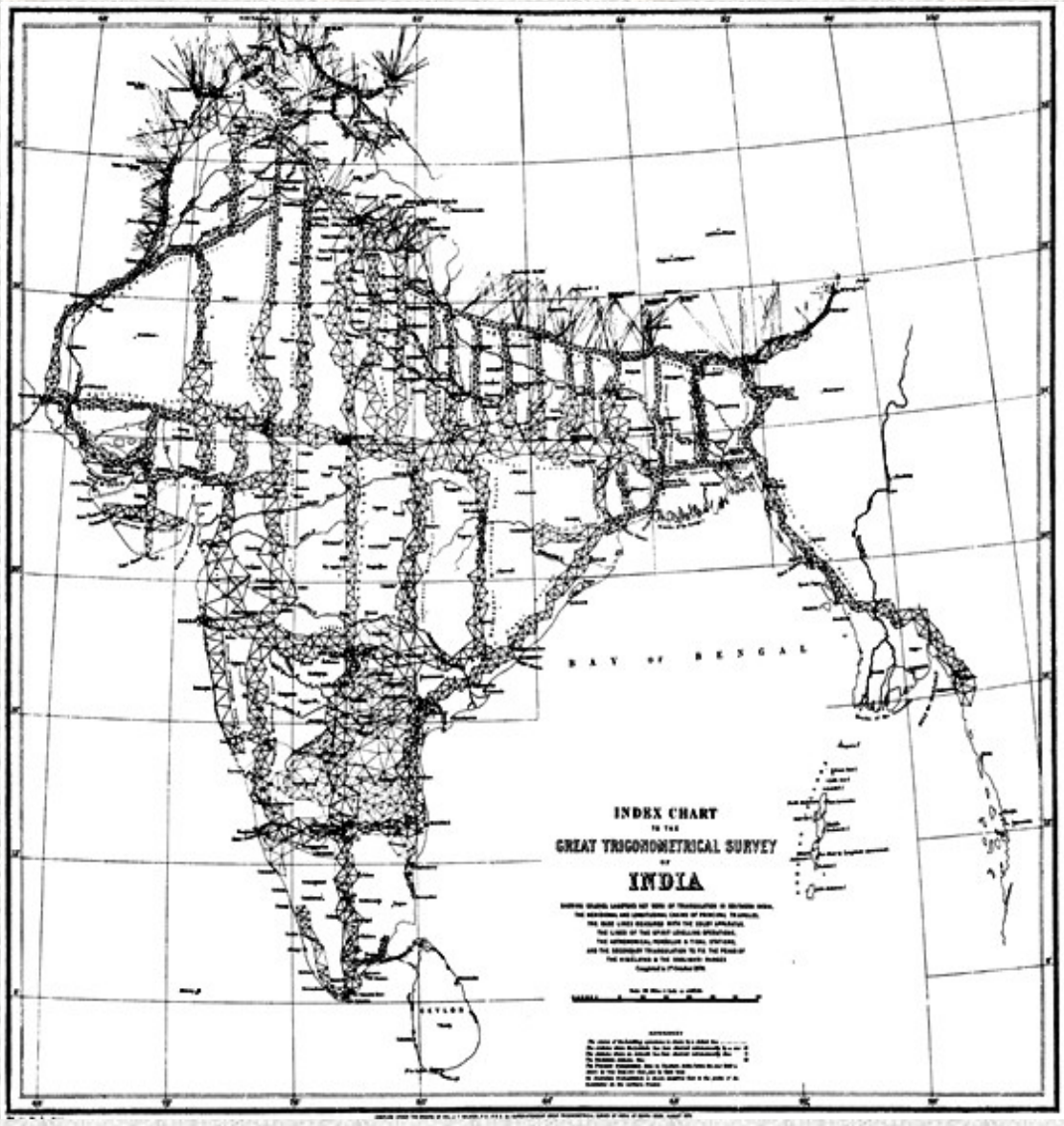
a. Triangulation
b. Traverse
c. Trilateration
d. Astronomical Determinations
e. Satellite Triangulation

a. TRIANGULATION

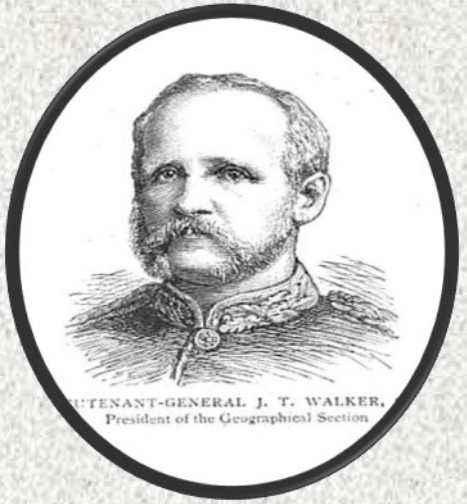
- It is one of the methods of providing control in an area which is to be surveyed



GREAT TRIGONOMETRIC SURVEY

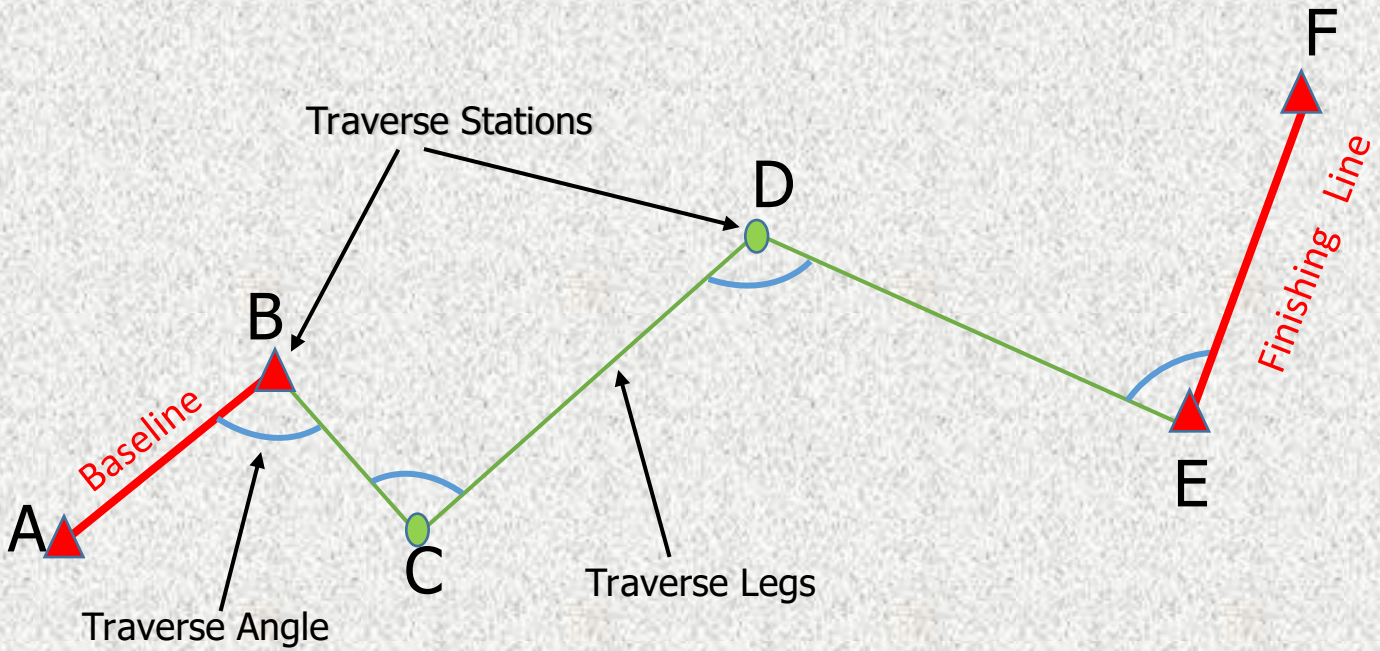


- **GREAT TRIGONOMETRIC SURVEY WAS COMPLETED IN 1882 BY GENERAL J.T WALKER**
- **IT PROVIDED A NETWORK OF SURVEY CONTROL POINTS IN THE SUBCONTINENT**



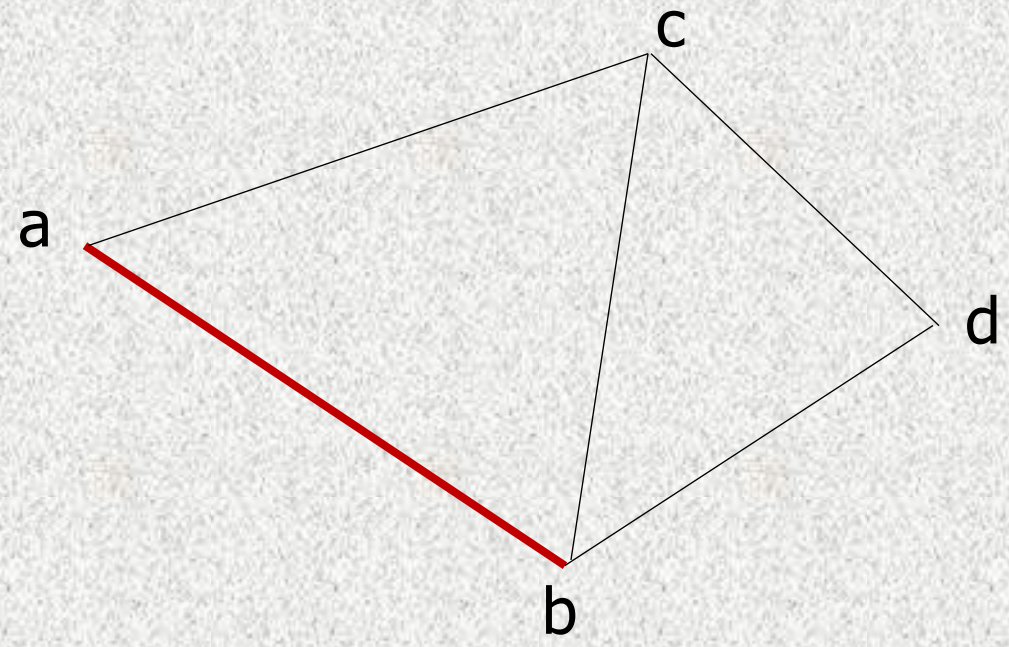
b. TRAVERSE

A traverse consists of a connected series of lines on earths' surface, the length and bearings of which have been determined.



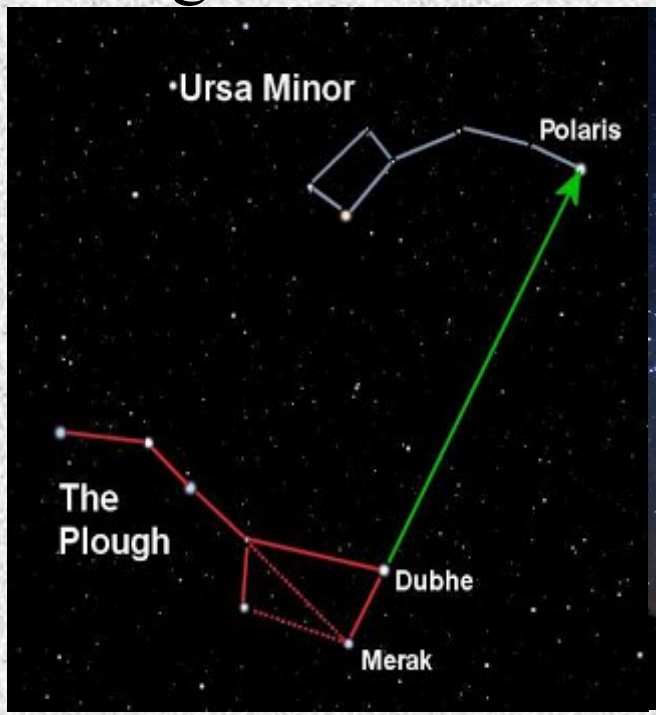
c. TRILATERATION

- Trilateration methods involve the determination of locations of controls points by measurement of distances, using the geometry of triangles. In contrast to triangulation it does not involve the measurement of angles.



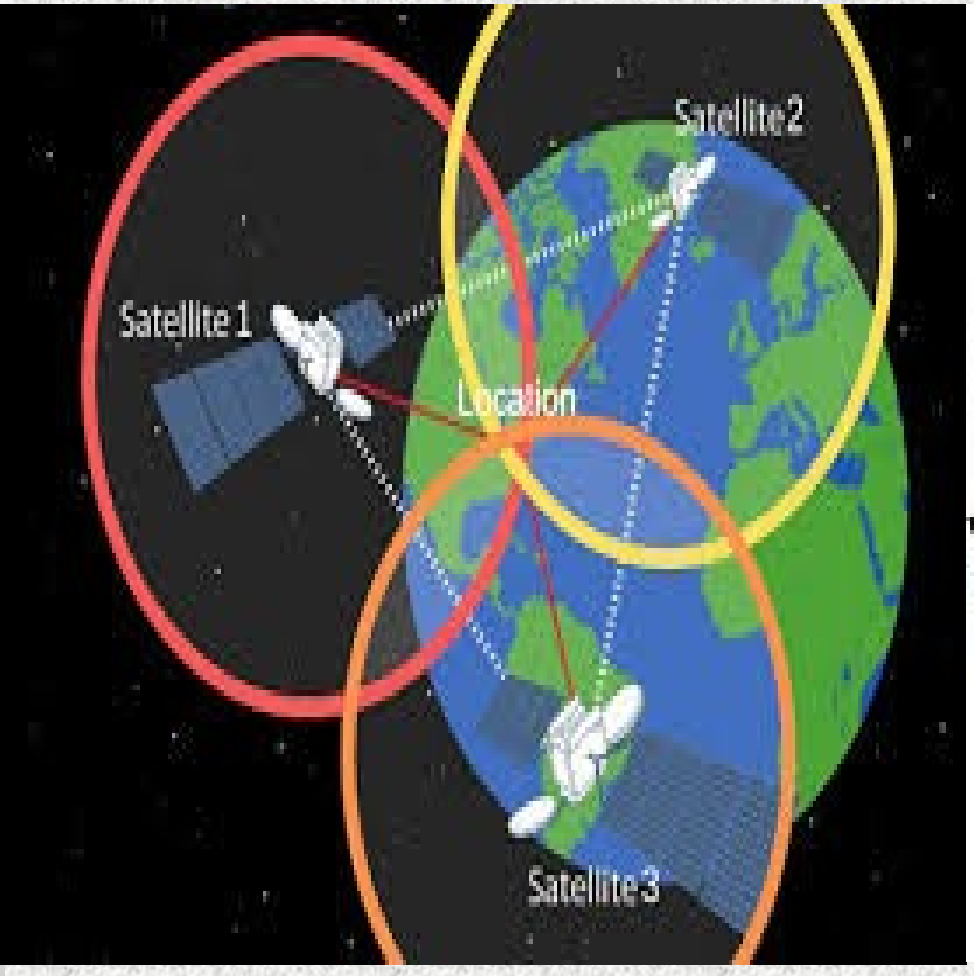
d. ASTRONOMICAL DETERMINATIONS

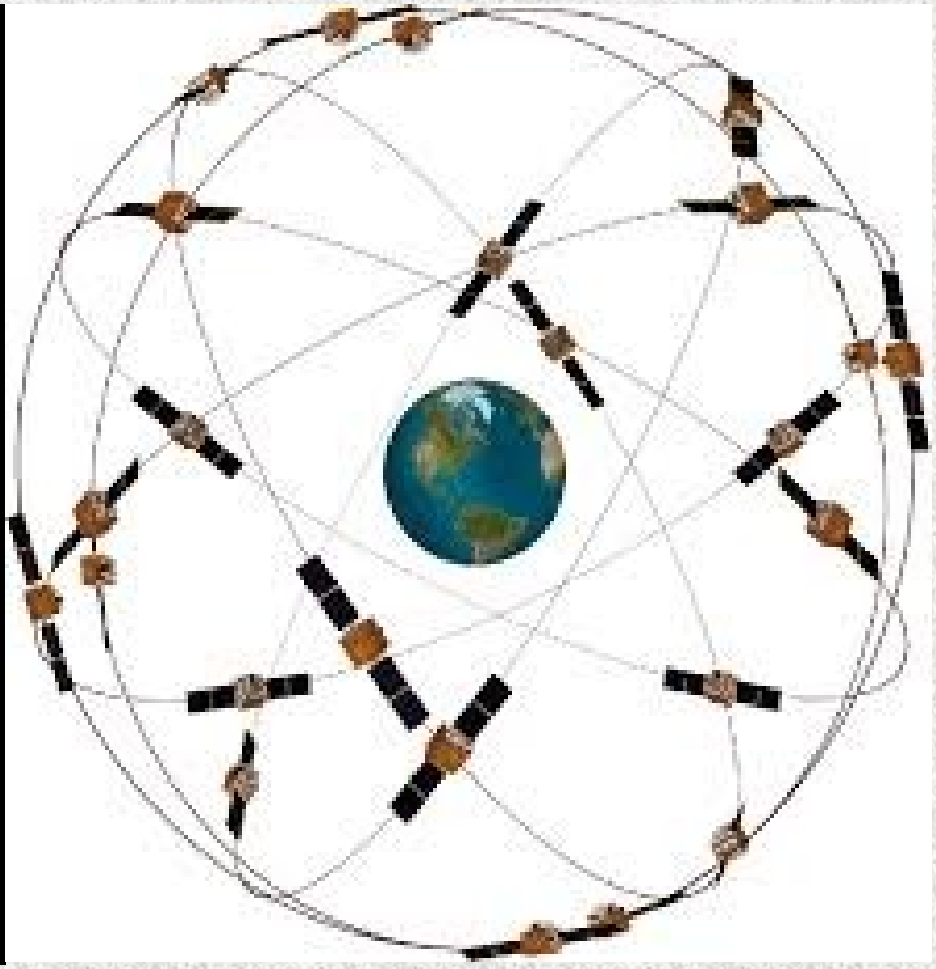
To determine position of a control point in terms of geographic coordinates i.e. longitudes and latitudes, using astronomical measurements





e. SATELLITE TRIANGULATIONS





PRINCIPALS OF SURVEY

2. **ECONOMY OF ACCURACY**
   - Greater accuracy requires greater time greater cost more refined methods.
3. **CONSISTANCY**
   - Same standard of accuracy to be maintained throughout the project.
4. **THE INDEPENDENT CHECK**
   - Use self checking methods
   - Use of different units of measurement
   - Measuring distance in reverse direction
   - Computations by another person, use different method

PRINCIPALS OF SURVEY

5. **REVISION**
   - Survey with revision in mind.
6. **SAFE GUARDING**
   - Easy for another person to understand.
   - Permanent marks as control points
   - Duplicate copies of field records
   - Soft copies of records

QUESTION